JEE Main

FST - 06

KEY SHEET

MATHEMATICS

1	**1**	2	**1**	3	**3**	4	**3**	5	**3**
6	**2**	7	**3**	8	**1**	9	**1**	10	**4**
11	**4**	12	**1**	13	**3**	14	**2**	15	**1**
16	**4**	17	**3**	18	**4**	19	**2**	20	**2**
21	**118**	22	**1012**	23	**60**	24	**12**	25	**2**

PHYSICS

26	**2**	27	**3**	28	**4**	29	**2**	30	**2**
31	**2**	32	**4**	33	**2**	34	**2**	35	**1**
36	**2**	37	**1**	38	**2**	39	**3**	40	**3**
41	**1**	42	**3**	43	**3**	44	**2**	45	**4**
46	**27**	47	**80**	48	**22**	49	**9**	50	**30**

CHEMISTRY

51	**3**	52	**1**	53	**1**	54	**3**	55	**1**
56	**3**	57	**2**	58	**1**	59	**4**	60	**4**
61	**3**	62	**1**	63	**3**	64	**1**	65	**3**
66	**4**	67	**3**	68	**4**	69	**4**	70	**3**
71	**240**	72	**21**	73	**59**	74	**16**	75	**4**

SOLUTION
MATHEMATICS

1. $\vec{v}\times\vec{w}=\vec{u}+\lambda\vec{v}$

$0=\vec{u}.\vec{v}+\lambda|\vec{v}|^2\Rightarrow\lambda=-\frac{1}{2}\qquad\therefore\ \vec{v}\times\vec{w}=\vec{u}-\frac{1}{2}\vec{v}$

$0=\vec{u}.\vec{w}-\frac{1}{2}\vec{v}.\vec{w}\text{ -}\quad\Rightarrow\vec{u}.\vec{w}=\frac{1}{2}\vec{v}.\vec{w}$

2. $y=mx+\frac{6}{m}\Rightarrow mx-y+\frac{6}{m}=0\ .....(1)$

$\frac{1}{2}(xy_1+yx_1)=2$

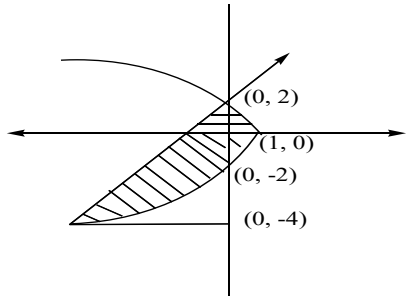
$xy_1+yx_1-4=0\ .....(2)$

$\frac{m}{y_1}=\frac{-1}{x_1}=\frac{-6}{4m}\qquad m=\frac{-y_1}{x_1}/\,m=\frac{3}{2}x_1\ \frac{-y_1}{x_1}=\frac{3}{2}x_1\Rightarrow y^2=-\frac{3}{2}x$

3. $\begin{vmatrix}1&1&1\\2&N&2\\3&3&N\end{vmatrix}\neq 0\qquad\Rightarrow N^2-6-1(2N-6)+1(6-3N)\neq 0$

$\Rightarrow N^2-5N+6\neq 0\Rightarrow N\neq 2,3$

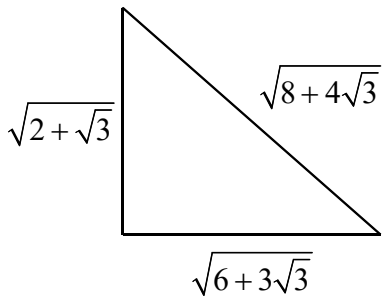
4. $y^2=-4(x-1);\ \ y=2x+2$



$y^2+2y-4-4=0\qquad y^2+2y-8=0\Rightarrow y=-4,2$

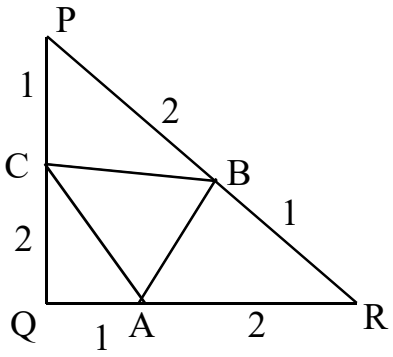
$\text{R.e A}=\int_{-4}^{2}\left(\frac{4-y^2}{4}-\frac{y-2}{2}\right)dy=9$

5.



$=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)+\tan^{-1}\sqrt{\frac{2+\sqrt{3}}{6+3\sqrt{3}}}\ =\tan^{-1}\frac{1}{\sqrt{3}}+\tan^{-1}\frac{1}{\sqrt{3}}=\frac{\pi}{3}$

6. Let $\overrightarrow{OQ} = \vec{r}$



$\overrightarrow{OR} = \vec{r}$

$\overrightarrow{OP} = \vec{p}$ -

$\overrightarrow{OA} = \dfrac{\vec{r}}{3}$

$\overrightarrow{OB} = \dfrac{\vec{p} + 2\vec{r}}{3}$

$\overrightarrow{OC} = 2\dfrac{\vec{p}}{3}$

Area of the triangle ABC = $\dfrac{1}{2}|\vec{p} \times \vec{r}|$

7. $\sum x_i = 10\ n \ldots\ldots(1)$

$\sum x_i - 8 + 12 = (10.2)\ n \ldots\ldots(2)$

$\sum x_i = (10.2)n - 4$

$10n = 10n + \dfrac{1}{5}n - 4 \qquad n = 20$

$\sum \dfrac{n_i^2}{20} - (10)^2 = 4 \quad \Rightarrow \sum x_i^2 = 2080$

$\dfrac{\sum x_i^2 - 8^2 + 12^2}{20} - (10.2)^2 = 3.96$

8. $\vec{a} = -\hat{i} + 2\hat{j} + \hat{k} = (-1, 2, 1)$

$\vec{a}, \vec{b}, \hat{j}$ are coplanar $\qquad \vec{b} = x.\vec{a} + y.\hat{j}$

Given $|\vec{b}| = \sqrt{6}$ and $\vec{a}.\vec{b} = 0$

9. $f'(x) = 8x^3 - 36x + 8 = 0 \Rightarrow 2x^3 - 9x + 2 = 0$

$\Rightarrow (x - 2)(2x^2 + 4x - 1) = 0 \qquad \Rightarrow x = \dfrac{-4 \pm \sqrt{16 + 8}}{2(2)}$

$x = -1 \pm \sqrt{\dfrac{3}{2}}$

We get max. at $x = -1 - \sqrt{\dfrac{3}{2}}$

10. Conceptual

11. $x^2+(y-2)^2=4\left(x^2+(y+1)^2\right)$

$\Rightarrow 3x^2+3y^2+12y=x^2+y^2+4y=0$

Centre = (0, -2)

12. $\sqrt{2}(\sin x-\cos x)\in[-2,2]$

$(m-2)+\sqrt{2}(\sin x-\cos x)\in[m-4,m]$

$m-4=1\Rightarrow m=5$

13. Conceptual

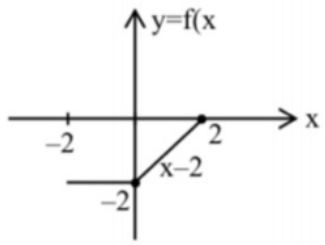
14. A B C

7R, 5B 5R, 7B 6R, 6B

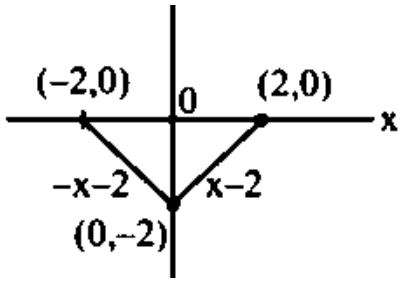
$P(B)=\frac{1}{3}.\frac{5}{12}+\frac{1}{3}.\frac{7}{12}+\frac{1}{3}.\frac{6}{12}$

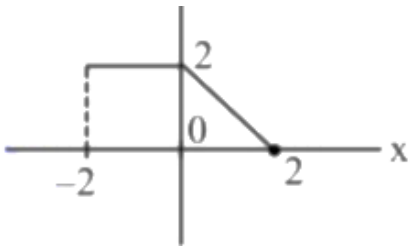
Required probability = $\frac{\frac{1}{3}.\frac{5}{12}}{\frac{1}{3}.\left[\frac{5}{12}+\frac{7}{12}+\frac{6}{12}\right]}=\frac{5}{18}$

15.

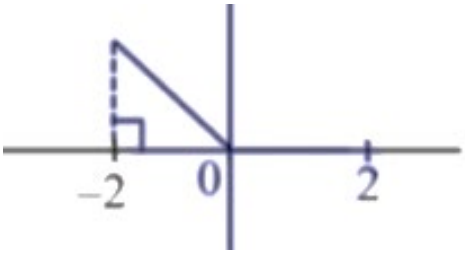


$f(|x|)\rightarrow$ $|f(x)|$



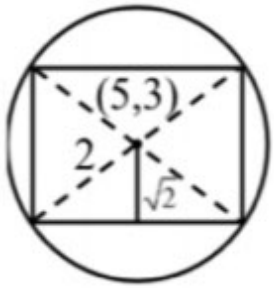


$$h(x)=\begin{cases}x-2+2-x=0, & 0\le x\le 2\\ -x-2+2=-x, & -2\le x<0\end{cases}$$



$=\int_0^2 h(x)dx=0$ and $\int_{-2}^0 h(x)dx=2$

16.



$y = x + c$ and $x + y + d = 0$

$\left|\frac{5-3+c}{\sqrt{2}}\right| = \sqrt{2}$ $\quad$ $\left|\frac{8+d}{\sqrt{2}}\right| = \sqrt{2}$

$|c+2| = 2$ $\quad$ $8 + d = \pm 2$

$c = 0, -4$ $\quad$ $d = -10, -6$

pts $(5, 5), (3, 3), (7, 3), (5,1)$

$\sum\left(x_i^2 + y_1^2\right) = 25 + 25 + 9 + 9 + 49 + 9 + 25 + 1 = 152$

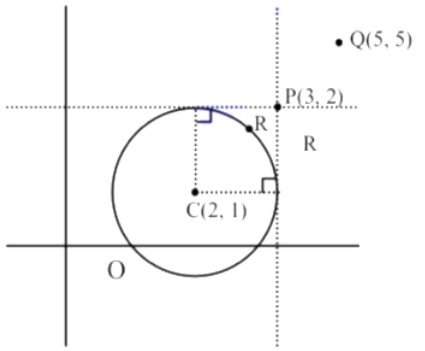
17. $ax^2 + bx + c = 0$ $\quad$ $a, b, c \in \{1, 2, 3, 4, 5, 6, 7, 8\}$

Repeated roots $D = 0$ $\quad$ $\Rightarrow b^2 - 4ac = 0 \Rightarrow b^2 = 4ac$

$\text{prob} = \frac{8}{8 \times 8 \times 8} = \frac{1}{64} \Rightarrow (a, b, c)$

(1, 2, 1); (2, 4, 2); (1, 4, 4); (4,

18.



19. Centre of the circle = $\left(\frac{3}{2}, 1\right)$

Equation of diameter = $2x + 3y - k = 0$

$2\left(\frac{3}{2}\right) + 3(1) - k = 0$ $\quad$ $\Rightarrow k = 6$

Now, equation of ellipse becomes

$x^2 + 9y^2 = 36$ $\quad$ $\frac{x^2}{6^2} + \frac{y^2}{2^2} = 1$

20. Conceptual

21. $(x-2)^2 + y^2 = r^2; \frac{x^2}{36} + \frac{y^2}{16} = 1$

$$(x-2)^2+16\left(1-\frac{x^2}{36}\right)=r^2$$

Discriminant = 0; we get 'r'

22. Use formula : $r.\ {}^{n}C_r = n.\ {}^{(n-1)}C_{r-1}$

23. 1 2 3 4 1 2 3 4 1

There 4 exam places and 4 excess digits

$$\frac{4!}{2!2!}\times\frac{5!}{3!2!}=6\times10=60$$

24. $a\left(\frac{1}{m}\right)^3=500\Rightarrow a=500\ m^3$

$$S_n-s_{n-1}=\frac{a\left(r^n-1\right)}{r-1}-\frac{a\left(r^{n-1}-1\right)}{r-1}=ar^{n-1}$$

$S_6-S_5>1$

$ar^5>1$

$500\ m^3\times\frac{1}{m^5}>1$

$\Rightarrow m^2<500\ .....(1)$

$m^3>10^3\Rightarrow m>10$

$\therefore m=11,12,13,.....12$

25. $\tan\left(\pi\cos\theta\right)+\tan\left(n\pi-\pi\sin\theta\right)=0$

$\Rightarrow\cos\theta-\sin\theta=n,\ n\ \text{is integer}$

$\cos\theta-\sin\theta\in\left[-\sqrt{2},\sqrt{2}\right]$

$\therefore n=-1,0,1$

$\cos\theta.\sin\theta=-1$

$$\cos\left(\theta+\frac{\pi}{4}\right)=-\frac{1}{\sqrt{2}}$$

$$\theta+\frac{\pi}{4}=\frac{3\pi}{4},\frac{5\pi}{4}$$

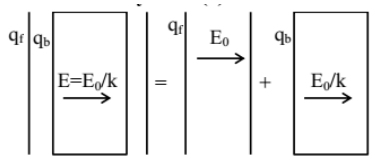
$$\cos\theta=\sin\theta\Rightarrow\theta=\frac{\pi}{4}$$

$$\cos\left(\theta+\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$$

$$\theta=\frac{7\pi}{4}$$

PHYSICS

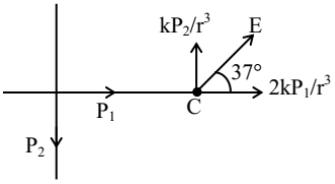
26.



When a dielectric is inserted in a capacitor due to free charge $\vec{E} = \vec{E}_0$ only

After dielectric $\vec{E} = \vec{E}_0$ only , After dielectric $E' = \dfrac{E_0}{k}$ $q_B = q_f\left(1 - \dfrac{1}{k}\right)$

27.



$$\tan 37^0 = \frac{3}{4} = \frac{\frac{kP_2}{r^3}}{\frac{2kP_1}{r^3}} = \frac{P_2}{2P_1} = \frac{3}{4} \qquad \frac{P_2}{P_1} = \frac{3}{2} \qquad \frac{P_1}{P_2} = \frac{2}{3}$$

28. $C_P - C_V = R$ for ideal gas and gas behaves as ideal gas at high temperature so $T_P > T_Q$

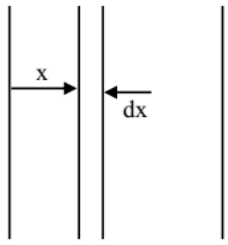
29. $I_z = I_x + I_y$ (using perpendicular axis theorem)

& $I = mk^2$ (K : radius of gyration)

so $mK_z^2 = mK_x^2 + mK_y^2$ $\quad K_z^2 = K_x^2 + K_y^2$

so radius of gyration ab out axes x, y & z won't be same hence Assertion (A) is not correct reason R is correct statement (property of a rigid body)

30.



Taking an element of width dx at a distance

x(x < d/2) from left plate $\qquad dc = \dfrac{(\varepsilon_0 + kx)A}{dx}$

Capacitance of half of the capacitor

$$\frac{1}{C} = \int_0^{d/2} \frac{1}{dc} = \frac{1}{A}\int_0^{d/2} \frac{dx}{\varepsilon_0 + kx} \quad \frac{1}{C} = \frac{1}{kA}\ln\left(\frac{\varepsilon_0 + kd/2}{\varepsilon_0}\right)$$

Capacitance of second half will be same

$$C_{eq} = \frac{C}{2} = \frac{kA}{2\ln\left(\frac{2\varepsilon_0 + kd}{2\varepsilon_0}\right)}$$

31. Fringe width, $\beta = \dfrac{\lambda D}{d}$

$\beta_{max} \Rightarrow d_{min}$ and $\beta_{min} \Rightarrow d_{max}$

$d = d_0 + a_0 \sin \omega t$

$d_{max} = d_0 + a_0$ and $d_{min} = d_0 - a_0$

$\therefore \beta_{min} = \dfrac{\lambda D}{d_0 + a_0}$ and $\therefore \beta_{max} = \dfrac{\lambda D}{d_0 - a_0}$

$$\beta_{max} - \beta_{min} = \frac{\lambda D}{d_0 - a_0} - \frac{\lambda D}{d_0 + a_0} = \frac{2\lambda D a_0}{d_0^2 - a_0^2}$$

32. Reflected wave will have direction opposite to incident wave.

33. Initially, energy of electron = +3eV

finally, in 2nd excited state

energy of electron = $-\dfrac{(13.6\ \text{eV})}{3^2} = -1.51\ \text{eV}$

Loss in energy is emitted as photon,

So, photon energy $\dfrac{hc}{\lambda} = 4.51\ \text{eV}$

Now, photoelectric effect equation

$$KE_{max} = \frac{hc}{\lambda} - \phi = 4.51 - \left(\frac{hc}{\lambda^{th}}\right) = 4.51\ \text{eV} - \frac{12400\ \text{eV}\,\overset{0}{\text{A}}}{4000\,\overset{0}{\text{A}}} = 1.41\ \text{eV}$$

34. Since, each vibrational mode, corresponds to two degrees of freedom, hence, f = 3 (trans) + 3 (rot) + 8 (vib.) = 14 & $\gamma = 1 + \dfrac{2}{f}$

$$\gamma = 1 + \frac{2}{14} = \frac{8}{7} \qquad W = \frac{nR\Delta T}{\gamma - 1} = -582$$

As W < 0, work done on the gas

35. q = CV

$$[C] = \left[\frac{q}{V}\right] = \frac{(A \times T)^2}{ML^2T^{-2}} = M^{-1}L^{-2}T^4A^2$$

$$[E] = \left[\frac{F}{q}\right] = \frac{MLT^{-2}}{AT} = MLT^{-3}A^{-1}$$

$$F = \frac{q_1 q_2}{4\pi \in_0 r^2}$$

$$[\in_0] = M^{-1}L^{-3}T^4A^2$$

Speed of light $c = \dfrac{1}{\sqrt{\mu_0 \in_0}}$ $\qquad \mu_0 = \dfrac{1}{\in_0 c^2}$

$$[\mu_0] = \frac{1}{\left[M^{-1}L^{-3}T^4A^2\right]\left[LT^{-1}\right]^2} = \left[M^{-1}L^1T^{-2}A^{-2}\right]$$

36. $E_{mech.} = 8J$

(A) at $x > x_4$, U = constant = 6 J

$K = E_{mech.} - U = 2$ J = constant

B) $x < x_1$, U = constant = 8 J

$K = E_{mech.} - U = 8 - 8 = 0$ J

Particle is at rest.

(C) At $x < x_2$, $U = 0 \Rightarrow K_{mech.} = K = 8J$

K.E is greatest, and particle is moving at fastest speed.

(D) At $x < x_3$, U = 4 J

U + K = 8 J

K = 4 J

37. $C < i_b$

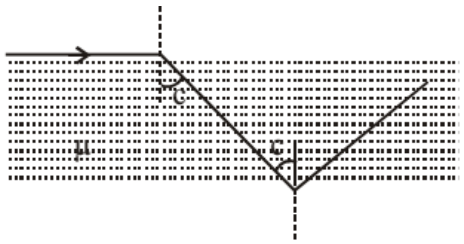
here i_b is 'brewester angle' and 'c' is critical angle

$$\sin_c < \sin i_b \quad \text{since } \tan i_b = \mu_{rel} = \frac{1.5}{\mu}$$

$$\frac{1}{\mu} < \frac{1.5}{\sqrt{\mu^2 + (1.5)^2}} \quad \therefore \sin i_b = \frac{1.5}{\sqrt{\mu^2 + (1.5)^2}}$$

$$\sqrt{\mu^2 \times (1.5)^2} < 1.5 \times \mu$$

$$\mu^2 \times (1.5)^2 < (\mu \times 1.5)^2 \qquad \mu < \frac{3}{\sqrt{5}}$$



slab $\mu = 1.5$

38. A) True, atom of each element emits characteristic spectrum.

(B) True, according to Bohr's postulates mvr = $\frac{nh}{2\pi}$ and hence electron resides into orbits of specific radius called stationary orbits.

(C) False, density of nucleus is constant

(D) False, A free neutron is unstable decays into proton and electron and antineutrino.

(E) True unstable nucleus show radioactivity

39. $t = 0, u = 0 \quad a = \frac{F_0}{M} - \frac{F_0}{MT^2}(t - T)^2 = \frac{dv}{dt}$

$$\int_0^v dv = \int_{t=0}^{2T} \left(\frac{F_0}{M} - \frac{F_0}{MT^2}(t - T)^2 \right) dt$$

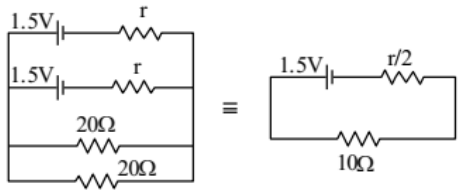
$$V = \left[\frac{F_0}{M} t \right]_0^{2T} - \frac{F_0}{MT^2} \left[\frac{t^3}{3} - t^2 T + T^2 t \right]_0^{2T} \quad V = \frac{4F_0 T}{3M}$$

40. Wattless component of current is I sin ϕ. Where ϕ is the phase angle

41. $\vec{F} = q\left(\vec{v} \times \vec{B}\right)$ $\quad \vec{F} \perp \vec{v}$

work done $= \vec{F}.\vec{S}$ work done = 0

42.



$$V = E - ir/2$$

$$1.2 = 1.5 - r\left(\frac{r}{2}\right) \qquad i = \frac{r}{2} = 0.3 \qquad i = \frac{1.5}{10 + \frac{r}{2}} \Rightarrow 10i + \frac{ir}{2} = 1.5$$

$$10i = 1.5 - 0.3 \qquad i = 0.12 \text{ A} \qquad \Rightarrow r = \frac{0.6}{0.12} = 5\Omega$$

43. $\mu_r = 1.61 \quad \in_r = 6.44$

$B = 4.5 \times 10^{-2}$

E = ?

$$C = \frac{1}{\sqrt{\mu \in_0}} V = \frac{1}{\sqrt{\mu \in}} \qquad \frac{C}{V} = \sqrt{\mu_r \in_r} = \sqrt{1.61 \times 6.44}$$

$$\frac{E}{B} = V = \frac{3 \times 10^8}{\sqrt{1.61 \times 6.44}} = 9.32 \times 10^7 \text{ m/s}$$

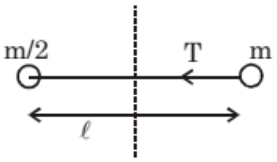
$$E = 4.5 \times 10^{-2} \times 9.32 \times 10^7 = 4.2 \times 10^6$$

44.

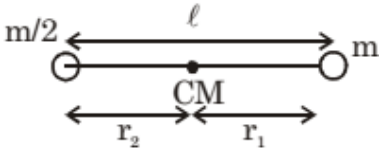
A	B	Y
1	1	0
0	0	1
0	1	1
1	0	1
1	1	0
0	0	1
0	1	1
1	0	1
NAND Gate		

$$Y = \overline{A.B}$$

45. $\omega = \sqrt{\frac{k}{I}}$



$$\omega = \sqrt{\frac{3k}{ml^2}} \quad = m\frac{3k}{ml^2}\theta_0^2\frac{l}{3} \qquad I = \mu l^2 \frac{\frac{m^2}{2}}{\frac{3m}{2}} l^2 \qquad = \frac{3l^2}{3}$$



$$\frac{r_1}{r_2} = \frac{1}{2} \Rightarrow r_1 = \frac{l}{3}$$

46. $\Delta m = \left(Zm_p + (A-Z)m_n\right) - M_{Al} \qquad = (13\times 1.00726 + 14\times 1.00866) - 27.18846$

$= 27.21562 - 27.18846 \quad = 0.02716\ u \qquad E = 27.15x\times 10^{-3} J$

47. $\phi = \vec{B}.\vec{S}$- $\qquad \phi = \frac{4}{\pi}\times 10^{-3}\left(1-\frac{t}{100}\right).\pi R^2$

$\phi = 4\times 10^{-3}\times (1)^2\left(1-\frac{t}{100}\right) \qquad \varepsilon = \frac{-d\phi}{dt}$

$\varepsilon = \frac{-d}{dt}\left(4\times 10^{-3}\left(1-\frac{t}{100}\right)\right) \qquad \varepsilon = 4\times 10^{-3}\left(\frac{1}{100}\right) = 4\times 10^{-5} V$

$1-\frac{t}{100} = 0 \quad t = 100\ sec \quad Heat = \frac{\left(4\times 10^{-5}\right)^2}{2\times 10^{-6}}\times 100\ J$

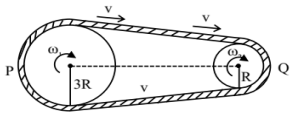
$Heat = \frac{16\times 10^{-10}\times 100}{2\times 10^{-6}}\ J \qquad Heat = 0.08 J \qquad Heat = 80 mJ$

48. $B = \mu.(H+I) \qquad B = \mu.H\left(1+\frac{1}{H}\right)$

$B = B_0(1+x) \qquad B - B_0 = B_0 x \qquad \frac{B-B_0}{B_0} = x$

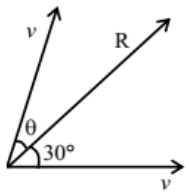
$\frac{B-B_0}{B_0}\times 100 = 100x \qquad = 2.2\times 10^{-3} = \frac{22}{10^4}$

49.



$\frac{1}{2}I_1(\omega_1)^2 = \frac{1}{2}I_2(\omega_2)^2 \qquad I_1\left(\frac{v}{3R}\right)^2 = I_2\left(\frac{v}{R}\right)^2 \qquad \frac{I_1}{I_2} = \frac{9}{1}$

50.



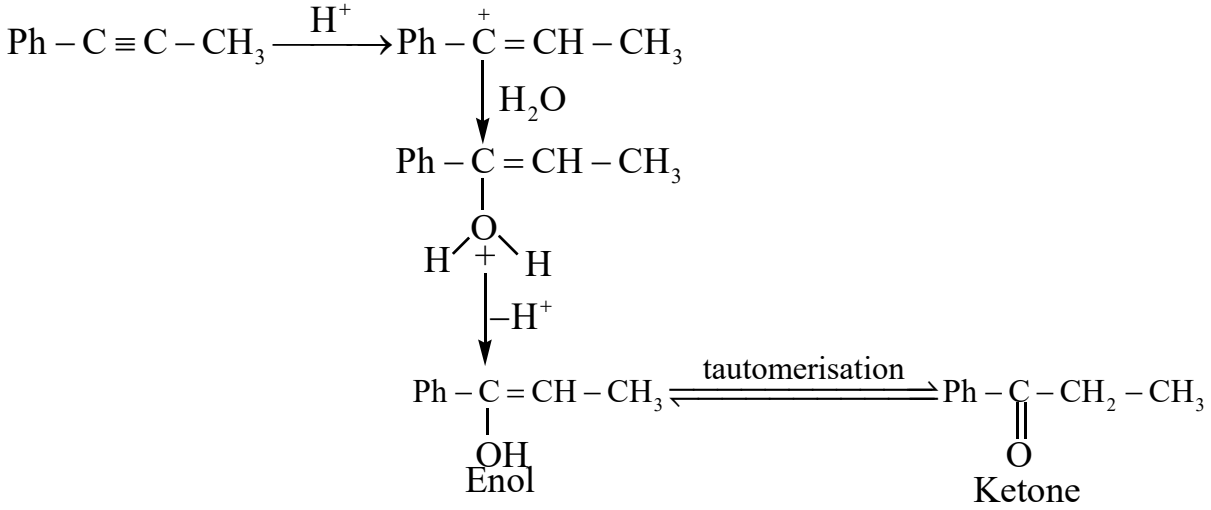
Both velocity vectors are of same magnitude therefore resultant would pass exactly midway through them $\theta = 30^0$

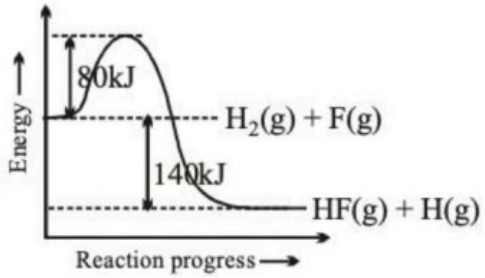
CHEMISTRY

51. → 2, 4 DNP test positive means Aldehyde (or) Ketone is present in compound.
→ Tollen's & Cannizaro Positive means Aromatic Aldehyde.
→ 1, 2 Dicarboxylic acids means ortho Isomer.

52. G.I is possible when compound with two same or different groups at atleast two sp^3 hybridized carbon atoms.

53. The given reaction represents the hydration of alkyne in acidic medium. Alkynes can be hydrated to form enols that immediately tautomerise to ketones.



54.


55. A) Phenol + Aniline - deviation from Raoult's law
B) Ethanol + Acetone -+ deviation from Raoult's law
C) n-hexane + n-heptane - Ideal solution
D) Osmotic pressure - -Colligative property

56. Statement -2 is incorrect because anomers are those which have difference in configuration at C-1.

57. It is characteristic of secondary amino familiar as Lieberman's nitroso amine test

58. $\ln K_p = \frac{-\Delta H}{R}\left(\frac{1}{T}\right) + \frac{\Delta S}{R}$

59. Less polar or non-polar compound elute first

60. Sulphur

61. O > C > B > N

62. $B_3N_3H_3$

63. The dissociation of water into H^+ and OH^- is an endothermic reaction.
$\therefore$ As temperature increase, its equilibrium constant also increases :
$\therefore \left[H^+\right]$ and $\left[OH^-\right]$ in water at equilibrium increases with increases in temperature.

$\therefore pH = -\log\left[H^+\right]$ decreases, as temperature increases.

So, both A and R are false.

65. $\Delta S_{system} = nR \ln\frac{p_1}{p_2} = 10 \times 8 \ln\frac{2}{1} = 56 \text{ J / K}$

$\Delta S_{surrounding} = -\frac{q}{T} \quad \Delta U = q + w = 0 \text{ or } q = -w = P_{ext}\left(V_2 - V_1\right)$

$q = P_{ext}\left(\frac{nRT}{P_2} - \frac{nRT}{P_1}\right) = nRT\left(\frac{P_{ext}}{P_2} - \frac{P_{ext}}{P_1}\right) = 10 \times 8 \times T\left(1 - \frac{1}{2}\right)$

$q = 40 \text{ T}$

$\Delta S_{surrounding} = \frac{-q}{T} \Rightarrow -40 \text{ J}; \quad \Delta S_{system} + \Delta S_{surr} = 56 - 40 \Rightarrow 16 \text{ J / K}$

66. $\frac{P^0 - P^s}{P^0} = \frac{n}{N} =$ (for dilute solution)

$\frac{0.5}{65.5} = \frac{n \times 18}{100}$

67. $C_x H_y + \left(x + \frac{y}{4}\right)O_2 \rightarrow x\,CO_2 + \frac{y}{2}H_2O$

Given that $x + \frac{y}{4} = 7.5$

$\frac{y}{2} = 3 \Rightarrow y = 6$

$x = 6$

i.e. C_6H_6

68. $r = a_0\frac{n^2}{Z} = a_0.\frac{(2)^2}{2} = 2a_0$

69. Conceptual

70. gm equivalent of Al deposited = $\frac{It}{96500}$

$\frac{w}{27} \times 3 = \frac{2 \times 30 \times 60}{96500}$

$w = 0.336 \text{ g}$

71. C.F.S.E $= -0.4\ \Delta_0$

$\Delta_0 = \frac{-192.9}{-0.4} = 480.5 \text{ KJ / mole}$

$\lambda = \frac{6.4 \times 10^{-34} \times 6 \times 10^{23} \times 3 \times 10^8}{480.5 \times 10^3}$

$\lambda = 0.239.7 \times 10^{-6}$

$\lambda = 240 \text{ nm}$

72. $K = \frac{2.303}{t} \log \frac{a}{a-x}$

$t = \frac{2.303}{20} \log 64 = \frac{2.303 \times 0.3 \times 6}{20}$

$t = 0.207 \text{ min}.$

$t = 0.21 \text{ min}.$

73. $M_{0.83}\ O_{1.00}$

$M_{83}\ 0_{100}$

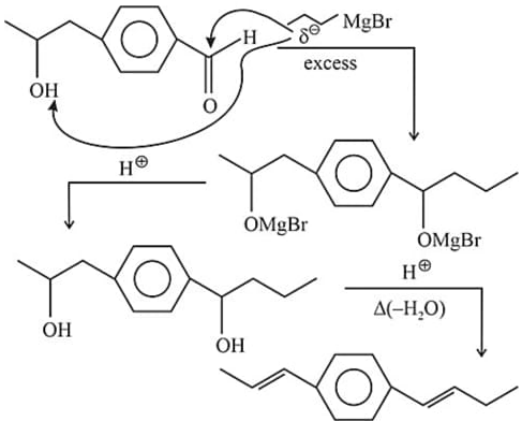
Let $+2$ of Mx, $+3$ of M $83 - x$

$2x + 3(83 - x) - 200 = 0$

$x = 49$

74. $K = \frac{K_2(K_3)}{K_1}$

75.



Here the product 'A' has two double bonds with different groups so it has 4 geometrical isomers.